Datasheet for Dataset

Neural Network for Payment Identity Resolution

1. Motivation

Dataset Name

Payment Identity Resolution Dataset (PIRD)

Purpose

This dataset was created to support the development of a machine learning and neural network solution capable of assigning payments made with a default taxpayer identification number (TIN) to the most probable registered taxpayer.

The dataset addresses challenges commonly encountered in:

- Fintech payment systems
- Revenue collection platforms
- Tax administration systems
- Banking reconciliation processes
- Government payment portals

Problem Statement

Many payment systems allow transactions to be processed using a default identifier such as:

```
9999999999
```

As a result:

- Payments cannot be accurately attributed
- Revenue becomes difficult to reconcile
- Audits become more complex
- Manual investigations increase

This dataset supports the training and evaluation of machine learning models that automate the identification process.

2. Dataset Composition

Data Sources

The dataset consists of:

Payment Records

Source:

```
March_Collections.xlsx
```

Contains payment transactions made by customers.

Taxpayer Records

Sources:

```
ITaxpayer1.xlsx
ITaxpayer2.xlsx
ITaxpayer3.xlsx
ITaxpayer4.xlsx
ITaxpayer5.xlsx
ITaxpayer6.xlsx
ITaxpayer7.xlsx
ITaxpayer8.xlsx
ITaxpayer9.xlsx
CTaxpayer.xlsx
```

Contains registered taxpayer information.

Dataset Size

Expected scale:

Dataset	Approximate Records
Collections	500,000+
Individual Taxpayers	1M+
Corporate Taxpayers	100K+

Actual size may vary depending on deployment.

Data Format

Primary format:

```
.xlsx
```

Supported exports:

```
.csv
.parquet
```

3. Data Schema

Payment Dataset

Field	Description
TIN	Taxpayer identifier
Payer Name	Name entered during payment
Phone Number	Payer phone number
Address	Payer address
Payment Reference	Unique transaction reference
Payment Amount	Amount paid
Payment Date	Date of transaction

Taxpayer Dataset

Field	Description
TIN	Official taxpayer identifier
Taxpayer Name	Registered name
Phone Number	Registered phone
Address	Registered address
Taxpayer Type	Individual or Corporate

4. Collection Process

Data Collection Method

Data is generated through:

- Online payment portals
- Tax collection systems
- Revenue management systems
- Banking integrations
- Fintech payment channels

Data Acquisition

Collected as part of normal business operations and taxpayer registration processes.

Time Period

Example:

```
March 2026
```

May be extended to additional months for retraining.

5. Preprocessing

Cleaning Activities

The following preprocessing steps are performed:

Name Standardization

Example:

```
JOHN DOE
John Doe
john doe
```

Converted to:

```
JOHN DOE
```

Address Normalization

Removes:

- Extra spaces

- Punctuation
- Special characters

Phone Number Standardization

Converts:

```
+2348031234567
08031234567
```

Into a common format.

Missing Values

Handled through:

- Null replacement
- Feature scoring adjustments
- Manual review flags

6. Labeling Process

Label Generation

Labels are generated using:

Rule-Based Matching

Examples:

- Exact phone match
- High name similarity
- Address similarity

Human Validation

Where available:

- Revenue officers
- Tax administrators
- Reconciliation teams

verify matches.

Label Definition

Label	Meaning
1	Correct Match
0	Not a Match

7. Dataset Splits

Recommended:

Split	Percentage
Training	70%
Validation	15%
Testing	15%

8. Uses

Intended Uses

- Payment attribution
- Taxpayer matching
- Revenue reconciliation
- Data quality improvement
- AI model development

Research Uses

- Entity resolution
- Record linkage
- Identity matching
- Similarity learning
- Fintech AI applications

9. Distribution

Access Level

Restricted/Internal

Contains potentially sensitive taxpayer information.

Storage

Recommended storage:

- Encrypted database
- Secure file repository
- Cloud object storage with RBAC

10. Privacy and Security

Sensitive Fields

The dataset may contain:

- Names
- Phone numbers
- Addresses
- Taxpayer IDs
- Payment references

Privacy Risks

Potential risks include:

- Identity exposure
- Unauthorized disclosure
- Re-identification attacks

Controls

Recommended controls:

- Data masking
- Encryption
- Access controls
- Audit logging
- Data retention policies

11. Bias and Fairness Considerations

Potential Sources of Bias

Data Quality Bias

Some taxpayers may have:

- Missing phone numbers
- Outdated addresses
- Misspelled names

Registration Bias

Corporate taxpayers may have more complete records than individuals.

Geographic Bias

Regions with lower-quality registration data may experience reduced matching accuracy.

Mitigation Strategies

- Regular data cleansing
- Human review workflow
- Confidence-based matching
- Periodic retraining

12. Maintenance

Update Frequency

Recommended:

- Monthly
- Quarterly

Retraining Trigger

Retraining should occur when:

- New taxpayer records exceed 10%
- Matching accuracy declines

- Data drift is detected

13. Dataset Relationships

Input Dataset

Payment transactions using default TINs.

Reference Dataset

Registered taxpayer master records.

Output Dataset

Enhanced payment records containing:

Field	Description
Probable TIN	Predicted taxpayer identifier
Confidence Score	Model confidence
Match Status	High, Medium, Low Confidence

14. Ethical Considerations

This dataset is intended to assist human decision-making and should not be used as the sole basis for:

- Legal determinations
- Tax enforcement actions
- Regulatory sanctions
- Customer blacklisting

Low-confidence predictions should always be reviewed by authorized personnel.

15. Citation

If using this dataset, cite as:

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Datasheet Version

Item	Value
Dataset Name	Payment Identity Resolution Dataset (PIRD)
Version	1.0
Date	2026
Status	Production/Research
Maintainer	Prercious Akpokighe
License	Internal/Restricted Use